

Assignment 6

Local testing:

Service returns 12 matching products in ~397µs locally, confirming bounded search checks exactly 100 products and returns up to 20 results.

```
(base) srinansiramesh@Srimansis-Laptop Assignments % curl "http://localhost:8080/products/search?q=alpha"

{"products":[{"id":8,"name":"Product Alpha 8","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":16,"name":"Product Alpha 16","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}, {"id":24,"name":"Product Alpha 24","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"}, {"id":32,"name":"Product Alpha 32","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"}, {"id":40,"name":"Product Alpha 40","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"}, {"id":48,"name":"Product Alpha 48","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":56,"name":"Product Alpha 56","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}, {"id":64,"name":"Product Alpha 64","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"}, {"id":72,"name":"Product Alpha 72","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"}, {"id":80,"name":"Product Alpha 80","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"}, {"id":88,"name":"Product Alpha 88","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":96,"name":"Product Alpha 96","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}], "total_found":12, "search_time":"397.583µs"}
```

Invalid query case:

Service correctly returns a structured `INVALID_INPUT` error when the required `q` parameter is missing.

```
(base) srinansiramesh@Srimansis-Laptop Assignments % curl "http://localhost:8080/products/search"

{"error":"INVALID_INPUT","message":"Missing search query","details":{"Query parameter 'q' is required"}}
```

AWS setup:

ECR:

Docker image successfully built and pushed to Amazon ECR repository `ecr_service` as `latest`.

ecr_service

Summary

Images

Lifecycle policy

Permissions

Repository tags

Images (3)

Info

Delete

Copy URI

Details

Scan

View push commands

Filter active images

☐	Image tags	Type	Created at	Image size	Image digest	Last pulled at
☐	latest	Image Index	22 February 2026, 17:42:14 (UTC-05)	14.02	sha256:31cccc5...	22 February 2026, 17:42:52 (UTC-05)
☐	-	Image	22 February 2026, 17:42:14 (UTC-05)	0.00	sha256:cdb60f...	22 February 2026, 17:42:14 (UTC-05)
☐	-	Image	22 February 2026, 17:42:14 (UTC-05)	14.02	sha256:5bcb2e...	22 February 2026, 17:42:52 (UTC-05)

ECS:

CS6650L2-cluster deployed on Fargate with 1 active service running on 0.25 vCPU / 512MB memory.

CS6650L2-cluster

Last updated

22 February 2026, 17:46 (UTC-5:00)

Actions

Create with Express Mode

Cluster overview

ARN

arn:aws:ecs:us-east-1:094522691730:cluster/CS6650L2-cluster

Status

Active

CloudWatch monitoring

Default

Registered container instances

-

Services

Draining

-

Active

1

Tasks

Pending

-

Running

1

Services

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

Tasks (4)

Last updated

22 February 2026, 17:46 (UTC-5:00)

Manage tags

Stop

Run new task

Filter tasks by property or value

Filter desired status

Any desired status

Filter launch type

Any launch type

< 1 >

☐	Task	Last status	Desired status	Task definition	Health status	Created at
☐	9d81045244754555ade7...	Running	Running	CS6650L2-task:3	Unknown	3 minutes ago
☐	80752ab272c64cfc8a9ac...	Stopped	Stopped	CS6650L2-task:3	Unknown	6 minutes ago
☐	fb410959f174465ca0991...	Stopped	Stopped	CS6650L2-task:3	Unknown	8 minutes ago
☐	fc926e7e4db34777b2bf5...	Running	Stopped	CS6650L2-task:3	Unknown	4 minutes ago

Testing:

Assignment 6

2

Service responds correctly from AWS with 103 μ s server-side search time, confirming deployment is healthy.

```
[(base) srinansiramesh@Srinansis-Laptop ~ % curl "http://100.24.62.239:8080/products/search?q=alpha"

{"products":[{"id":8,"name":"Product Alpha 8","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":16,"name":"Product Alpha 16","category":"Clothing","description":"Best seller in its category","brand":"Alpha"},{"id":24,"name":"Product Alpha 24","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"},{"id":32,"name":"Product Alpha 32","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"},{"id":40,"name":"Product Alpha 40","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"},{"id":48,"name":"Product Alpha 48","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":56,"name":"Product Alpha 56","category":"Clothing","description":"Best seller in its category","brand":"Alpha"},{"id":64,"name":"Product Alpha 64","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"},{"id":72,"name":"Product Alpha 72","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"},{"id":80,"name":"Product Alpha 80","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"},{"id":88,"name":"Product Alpha 88","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":96,"name":"Product Alpha 96","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}],"total_found":12,"search_time":"103.697us"}
```

Testing with 1 user:

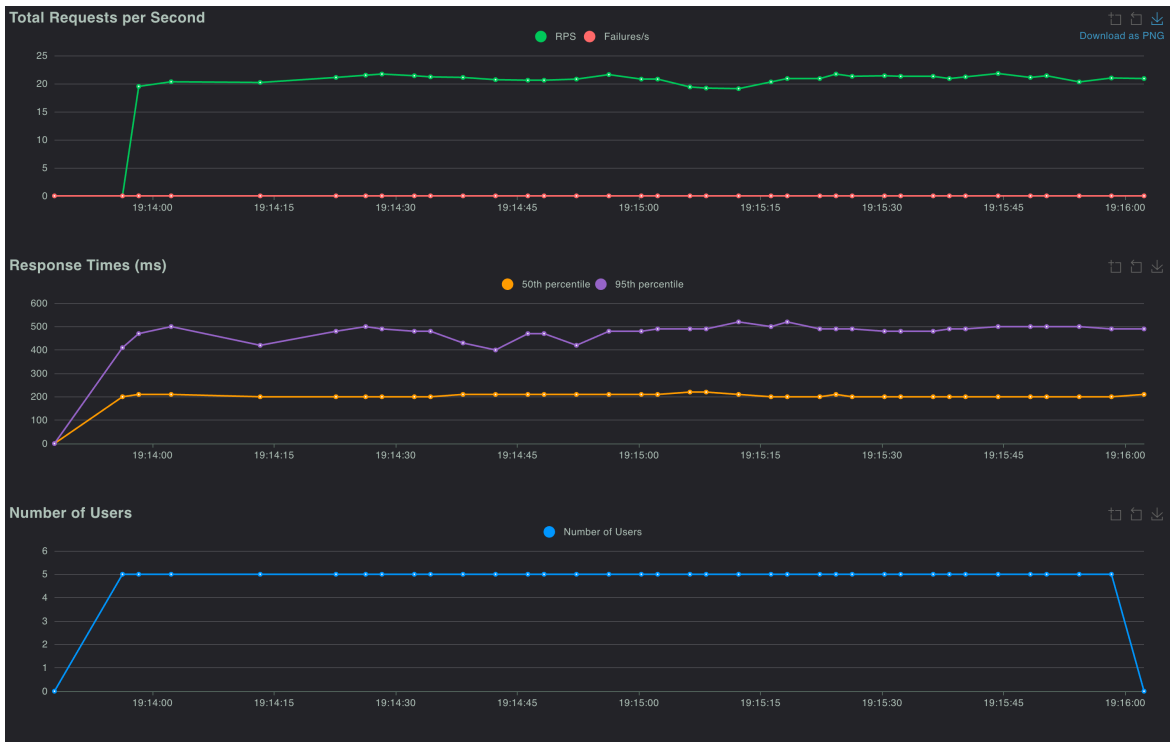
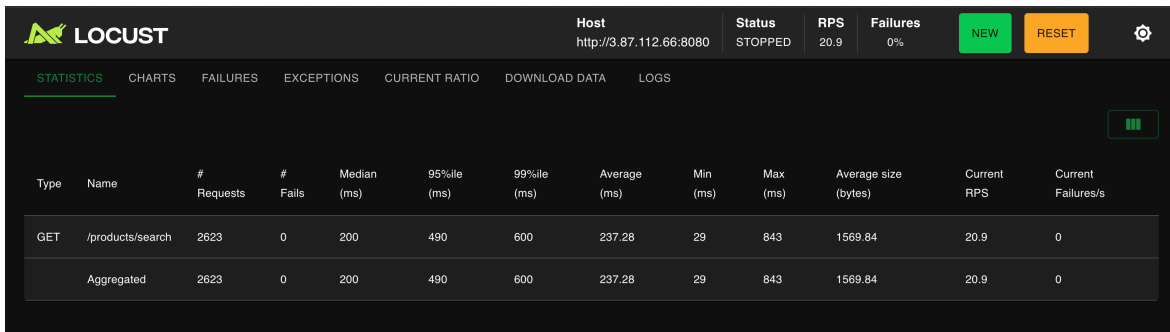
Single request completes in 133ms end-to-end. SHA-256 simulation adds realistic CPU load per request.

```
[(base) srinansiramesh@Srinansis-Laptop locust-tests % curl -w "\nTime: %{time_total}s\n" "http://100.24.62.239:8080/products/search?q=alpha"

{"products":[{"id":8,"name":"Product Alpha 8","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":16,"name":"Product Alpha 16","category":"Clothing","description":"Best seller in its category","brand":"Alpha"},{"id":24,"name":"Product Alpha 24","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"},{"id":32,"name":"Product Alpha 32","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"},{"id":40,"name":"Product Alpha 40","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"},{"id":48,"name":"Product Alpha 48","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":56,"name":"Product Alpha 56","category":"Clothing","description":"Best seller in its category","brand":"Alpha"},{"id":64,"name":"Product Alpha 64","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"},{"id":72,"name":"Product Alpha 72","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"},{"id":80,"name":"Product Alpha 80","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"},{"id":88,"name":"Product Alpha 88","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"},{"id":96,"name":"Product Alpha 96","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}],"total_found":12,"search_time":"75.588us"}
Time: 0.133765s
```

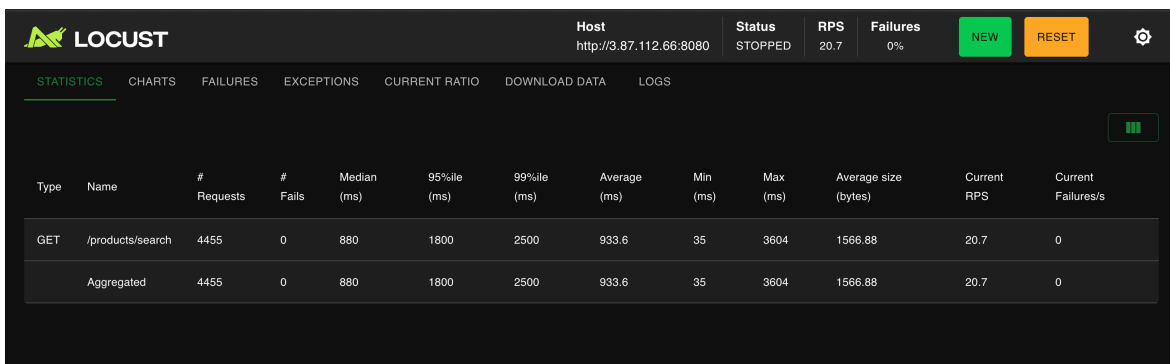
Testing with 5 users and 5 spawn rate:

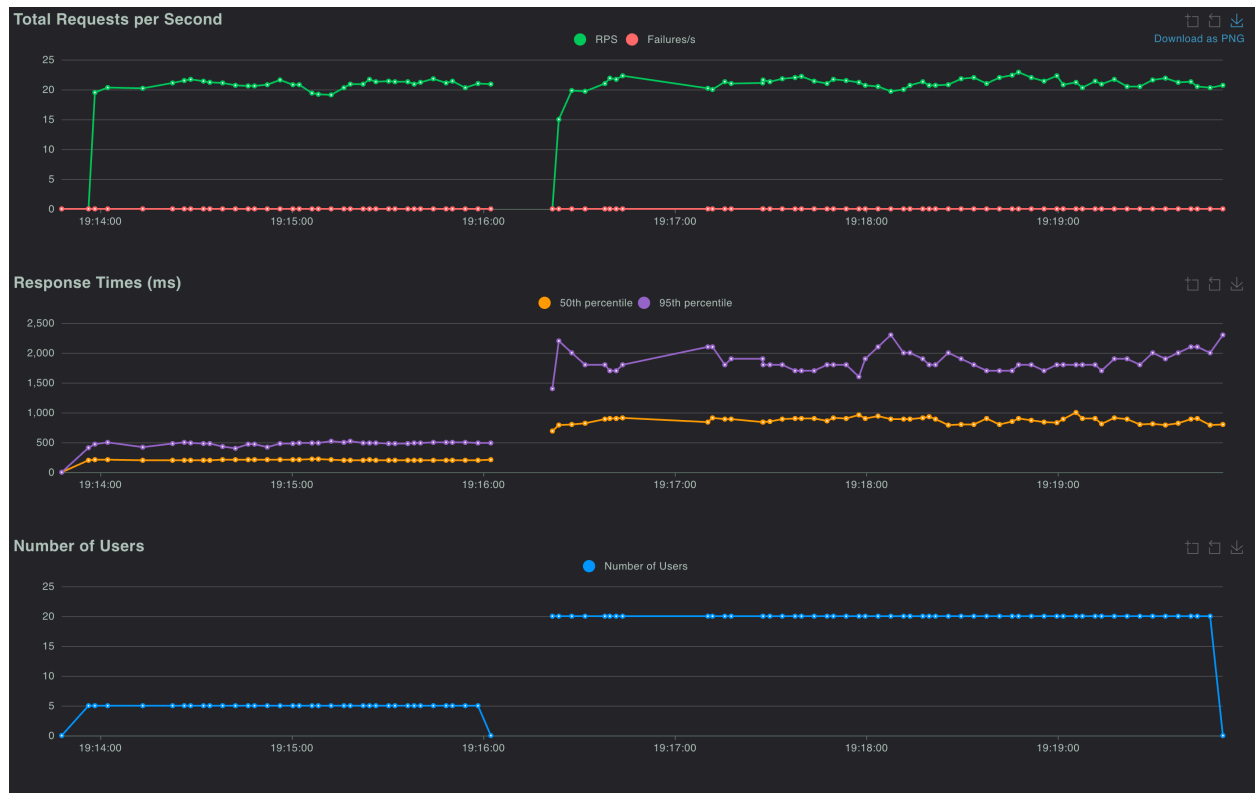
RPS-20.9; Median- 200ms; 95th percentile-490ms; 99th percentile- 600ms



Testing with 20 users and 20 spawn rate:

RPS- 20.7ms; Median- 880ms; 95th Percentile- 1800ms; 99th Percentile- 2500ms





Testing with 50 users and 50 spawn rate:

RPS- 21.4; Median- 2100ms; 95th Percentile- 5000ms; 99th Percentile - 6800ms

 LOCUST

Host

http://3.87.112.66:8080

Status

STOPPED

RPS

21.4

Failures

0%

NEW

RESET



STATISTICS

CHARTS

FAILURES

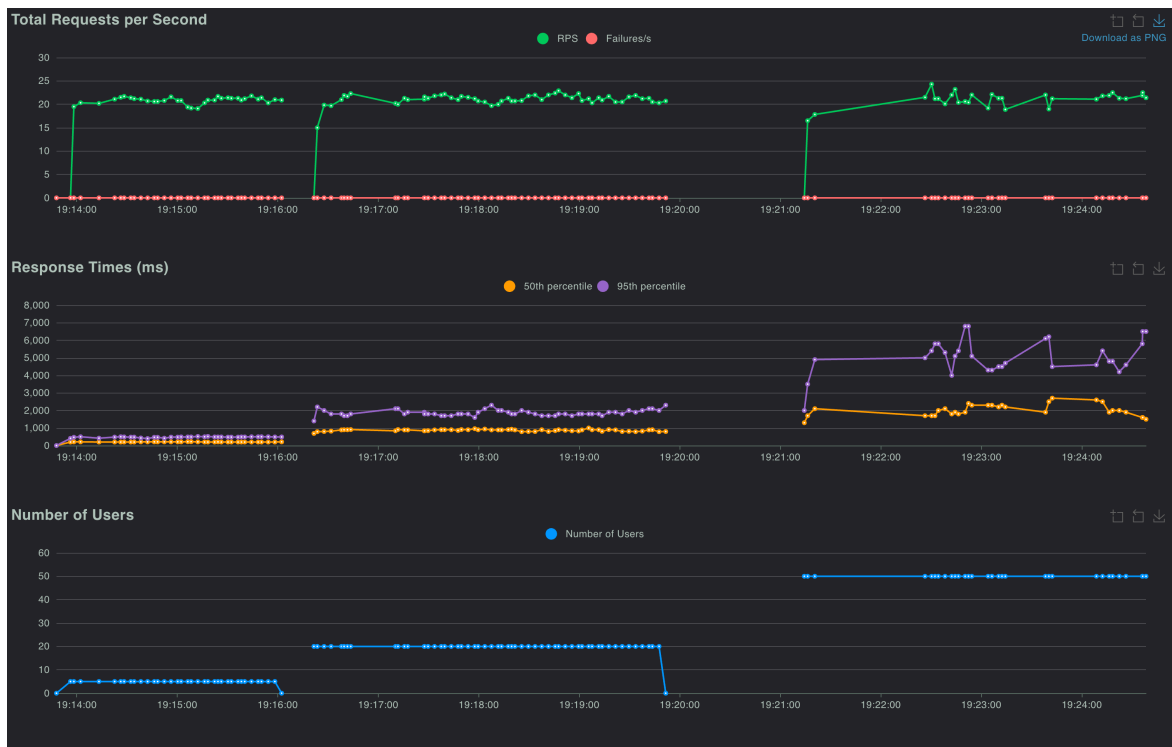
EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search	4401	0	2100	5000	6800	2309.23	46	10303	1579.83	21.4	0
	Aggregated	4401	0	2100	5000	6800	2309.23	46	10303	1579.83	21.4	0



Testing with 100 users and 100 spawn rate:

RPS- 22; Median- 4000ms; 95th Percentile - 12000ms; 99th Percentile- 15000ms

 LOCUST

Host

http://3.87.112.66:8080

Status

STOPPED

RPS

22

Failures

0%

NEW

RESET



STATISTICS

CHARTS

FAILURES

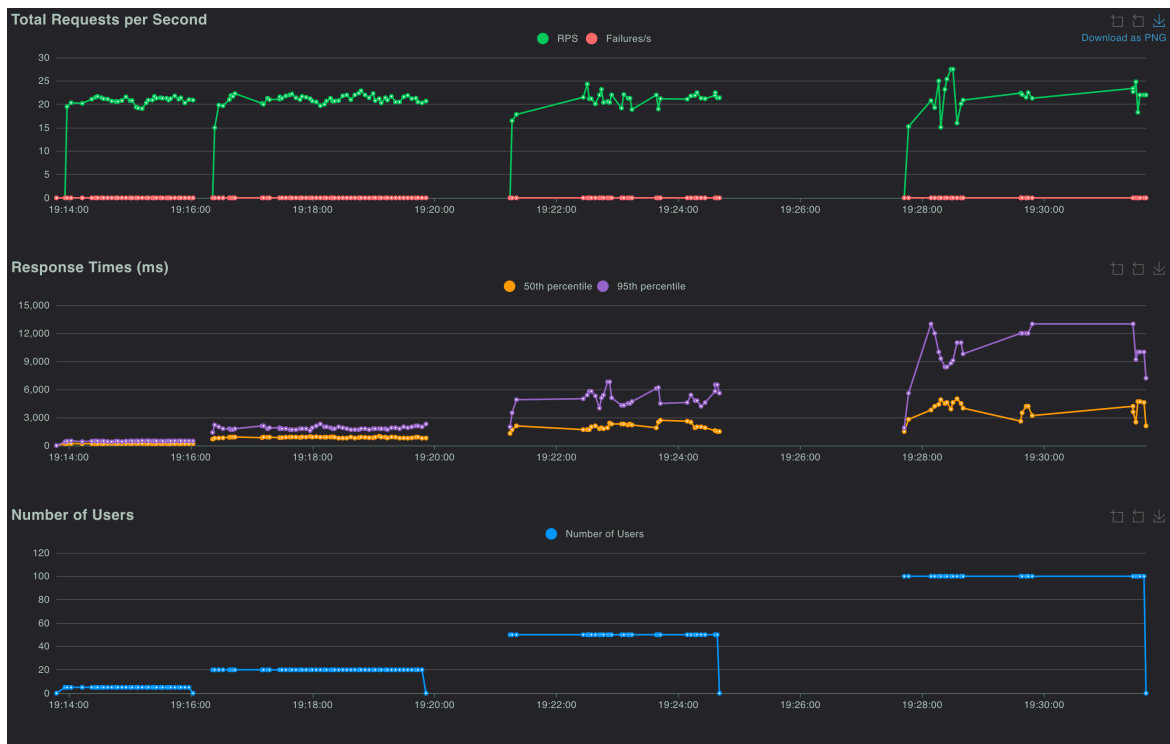
EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

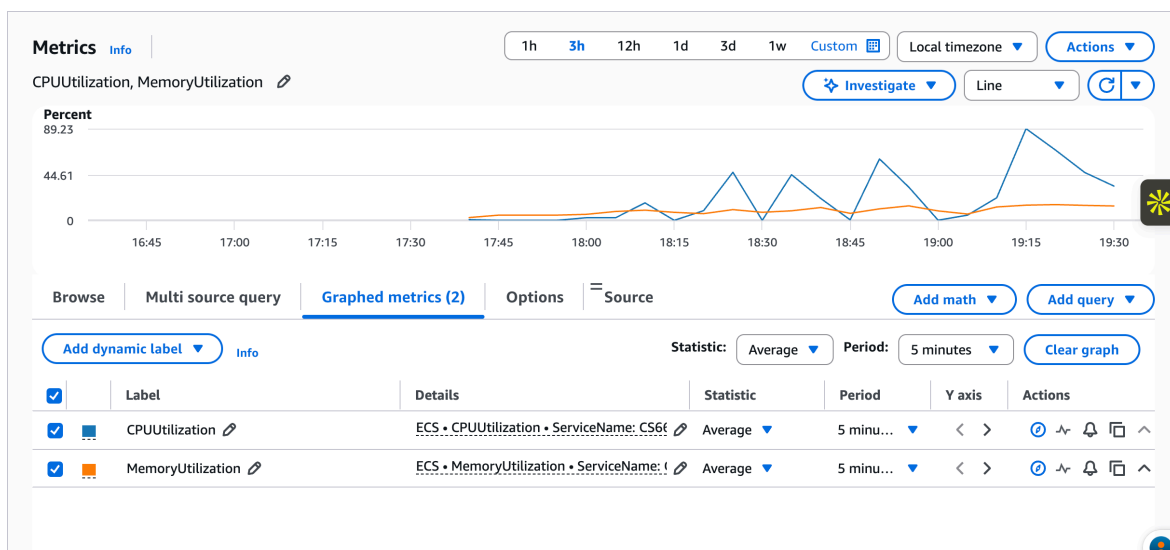
LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search	5065	0	4000	12000	15000	4631.82	41	25997	1561.82	22	0
	Aggregated	5065	0	4000	12000	15000	4631.82	41	25997	1561.82	22	0



Memory and CPU utilization:

CPU peaked at **89%** while memory stayed flat at ~44%. The fix is more compute (horizontal scaling), not code optimization.



Part3- Load balancer setup

CS6650L2-alb deployed as internet-facing Application Load Balancer, forwarding HTTP:80 to target group **CS6650L2-tg** across 6 availability zones with health checks at **/health**.

The screenshot displays the AWS Management Console interface for the 'Load balancers' section. At the top, there's a header with 'Load balancers (1/1)' and a 'What's new?' link. Below this, a sub-header states: 'Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.' A search bar labeled 'Filter load balancers' is present. A table lists the load balancers, with 'CS6650L2-alb' selected. The table columns include Name, State (Active), Type (application), Scheme (Internet-facing), IP address type (IPv4), VPC ID (vpc-05b6f59630c3b4e68), and Availability (6 Availability Zones). Below the table, the 'Load balancer: CS6650L2-alb' details are shown. The 'Listeners and rules' tab is active, displaying a table with one listener: 'HTTP:80'. The listener's default action is 'Forward to target group CS6650L2-tg: 1 (100%)'. The target group stickiness is set to 'Off'. The table columns include Protocol:Port, Default action, Rules, ARN, Security policy, and Default SSL.

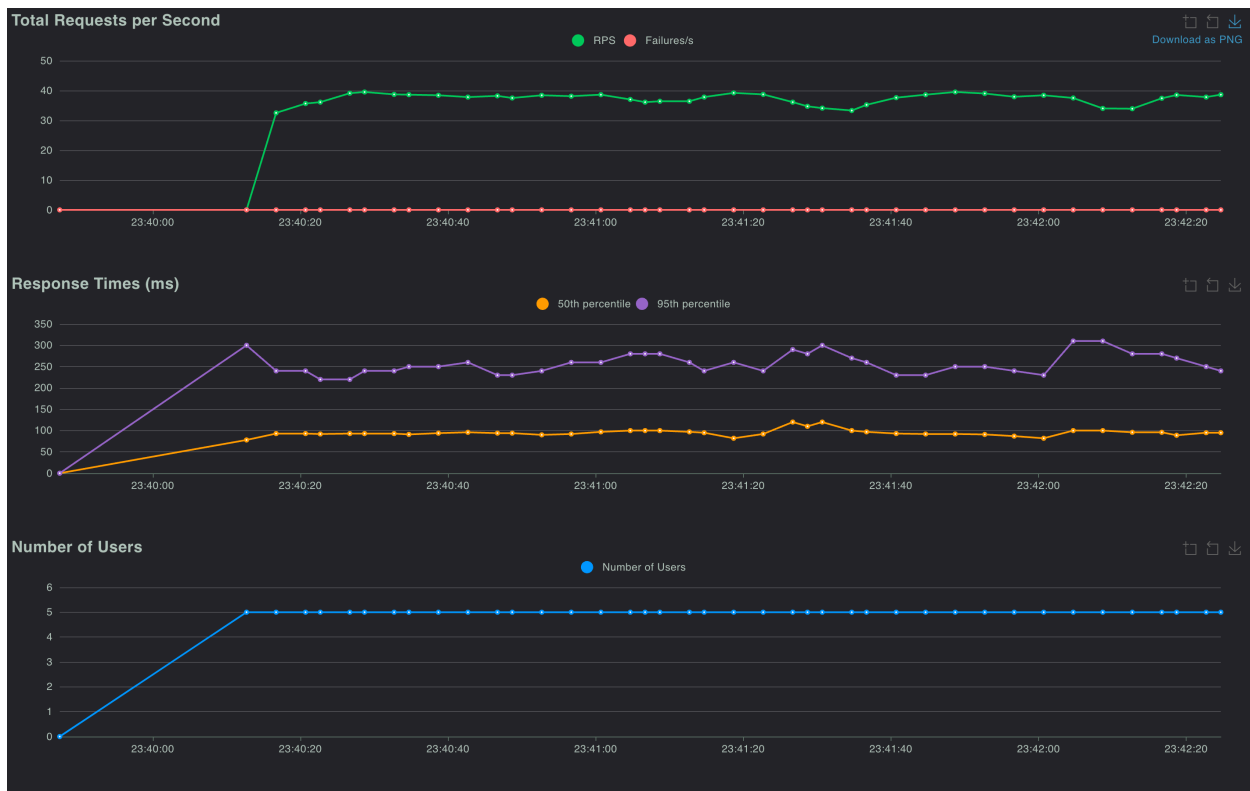
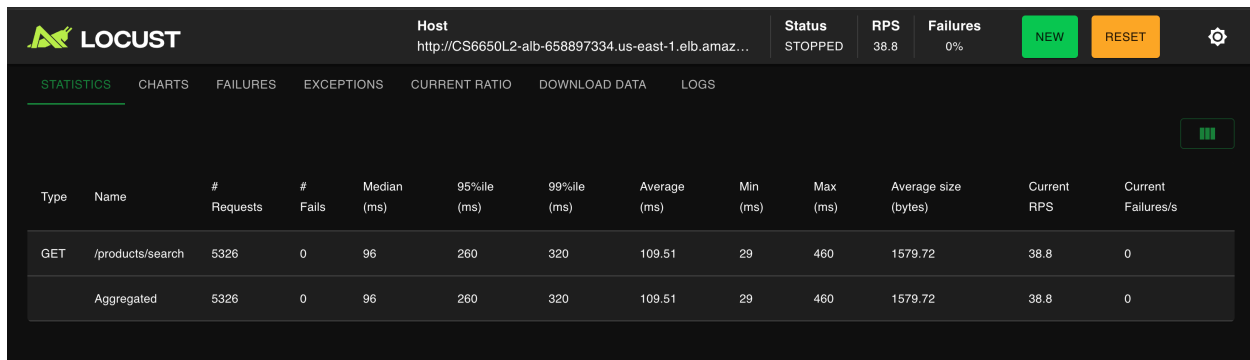
Testing ALB:

```
(base) srinansiramesh@Srimansis-Laptop src % curl http://CS6650L2-alb-658897334.us-east-1.elb.amazonaws.com/health
{"status":"healthy"}
(base) srinansiramesh@Srimansis-Laptop src % curl "http://CS6650L2-alb-658897334.us-east-1.elb.amazonaws.com/products/search?q=alpha"
{"products":[{"id":8,"name":"Product Alpha 8","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":16,"name":"Product Alpha 16","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}, {"id":24,"name":"Product Alpha 24","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"}, {"id":32,"name":"Product Alpha 32","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"}, {"id":40,"name":"Product Alpha 40","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"}, {"id":48,"name":"Product Alpha 48","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":56,"name":"Product Alpha 56","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}, {"id":64,"name":"Product Alpha 64","category":"Sports","description":"Top rated by customers worldwide","brand":"Alpha"}, {"id":72,"name":"Product Alpha 72","category":"Home","description":"Premium grade with warranty included","brand":"Alpha"}, {"id":80,"name":"Product Alpha 80","category":"Electronics","description":"High quality product with excellent features","brand":"Alpha"}, {"id":88,"name":"Product Alpha 88","category":"Health","description":"Affordable and reliable everyday item","brand":"Alpha"}, {"id":96,"name":"Product Alpha 96","category":"Clothing","description":"Best seller in its category","brand":"Alpha"}], "total_found":12, "search_time":"9.186234ms"}
```

Part 3 testing

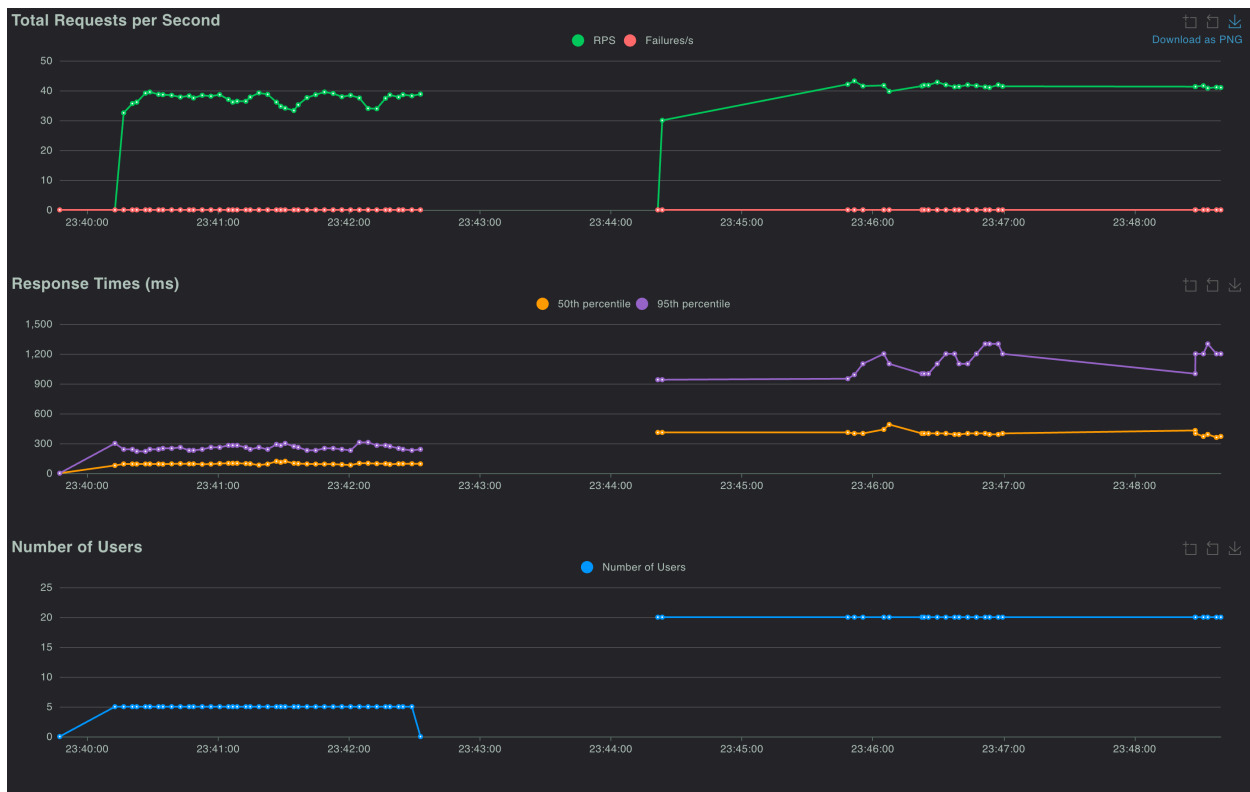
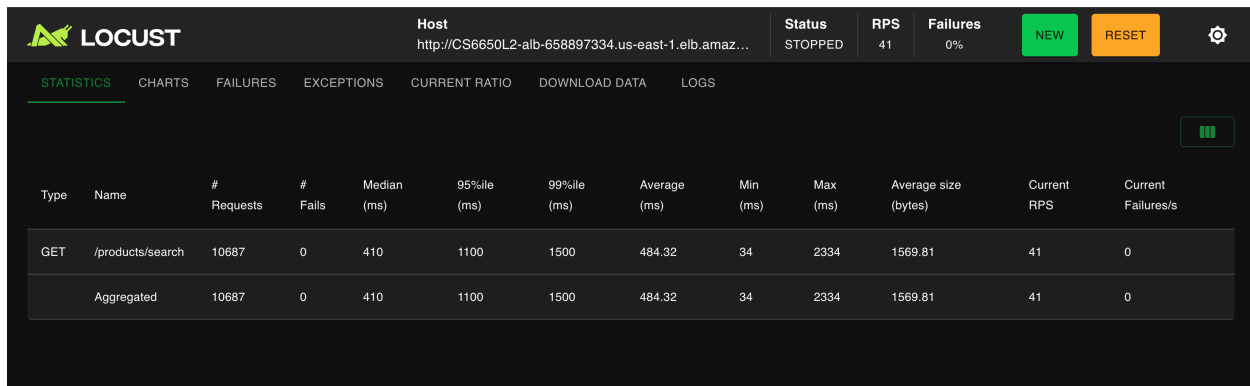
5 users 5 spawn rate

RPS- 38.8; Median- 96ms; 95th Percentile- 260ms; 99th Percentile- 320ms



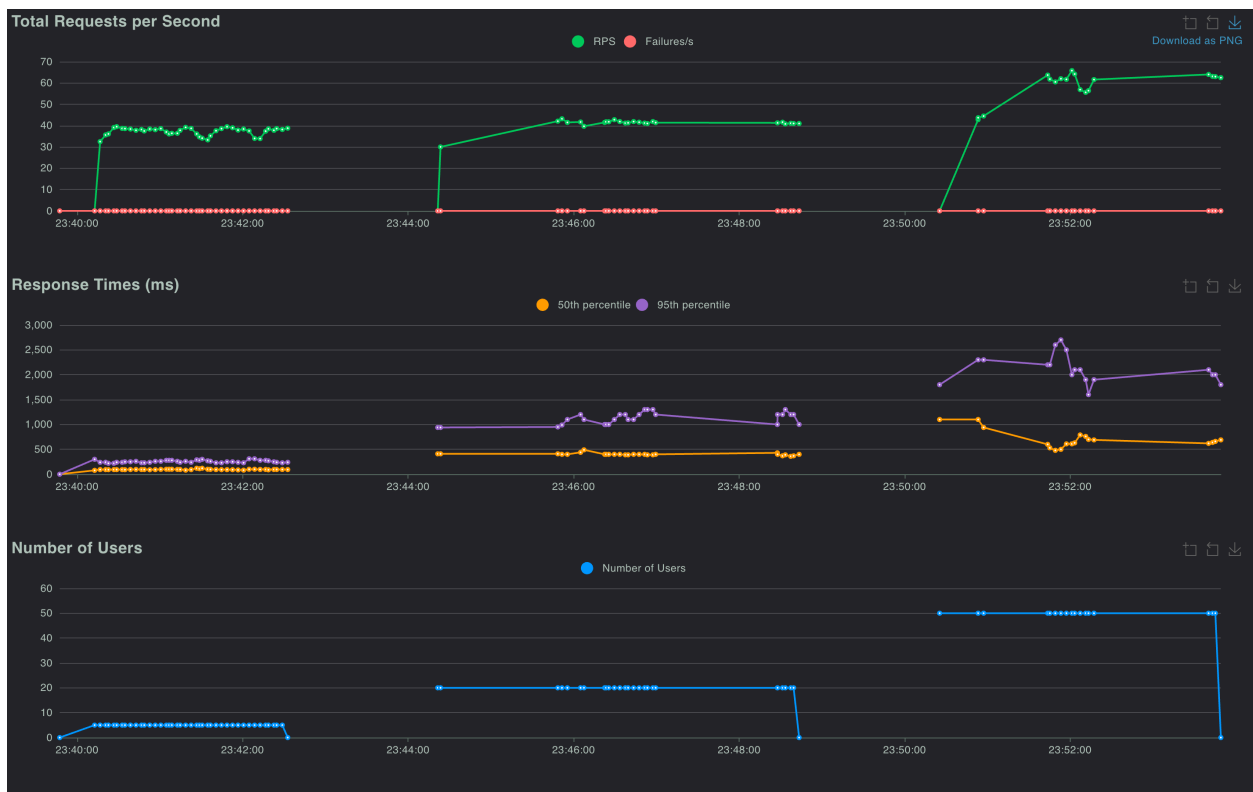
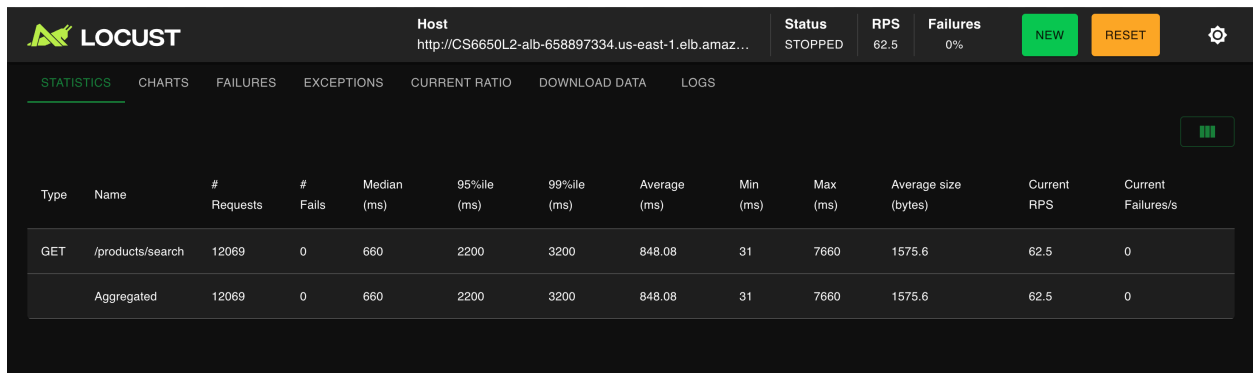
20 users and 20 spawn rate:

RPS- 41; Median- 410ms; 95th Percentile- 1100ms; 99th Percentile- 1500ms



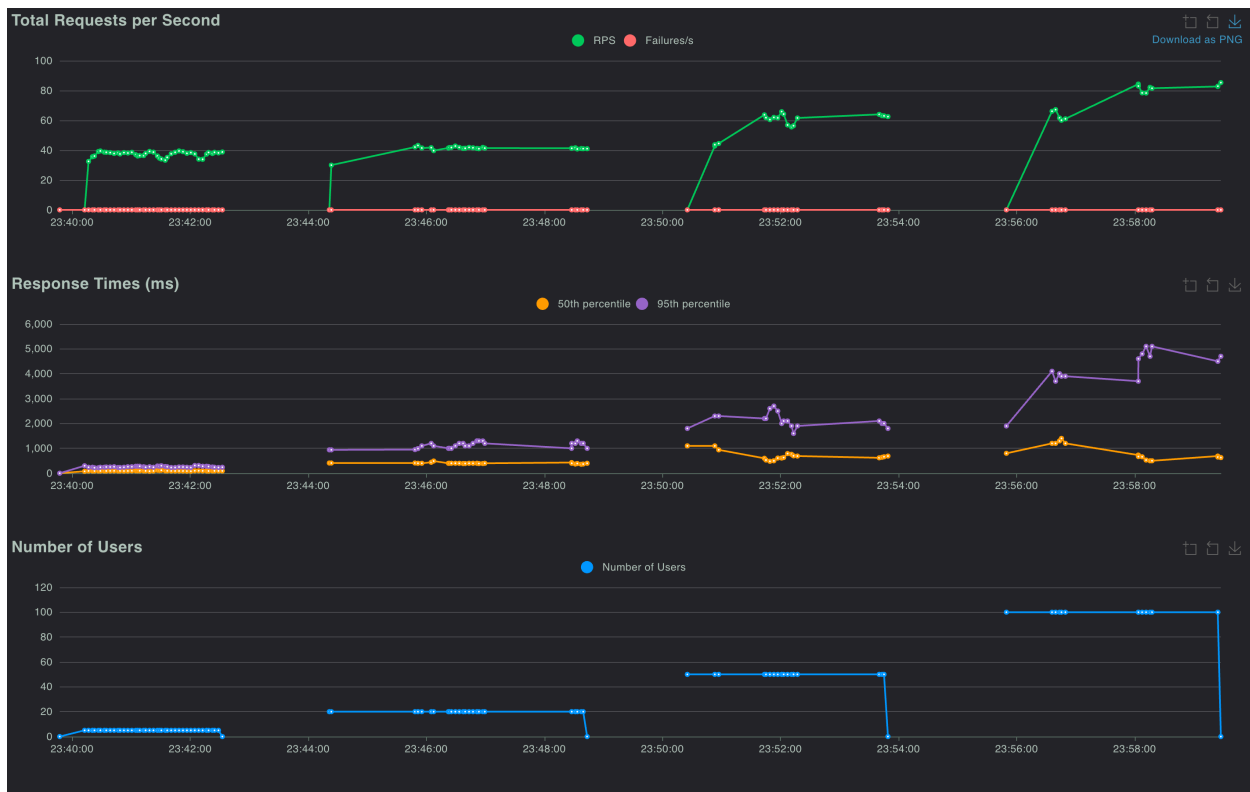
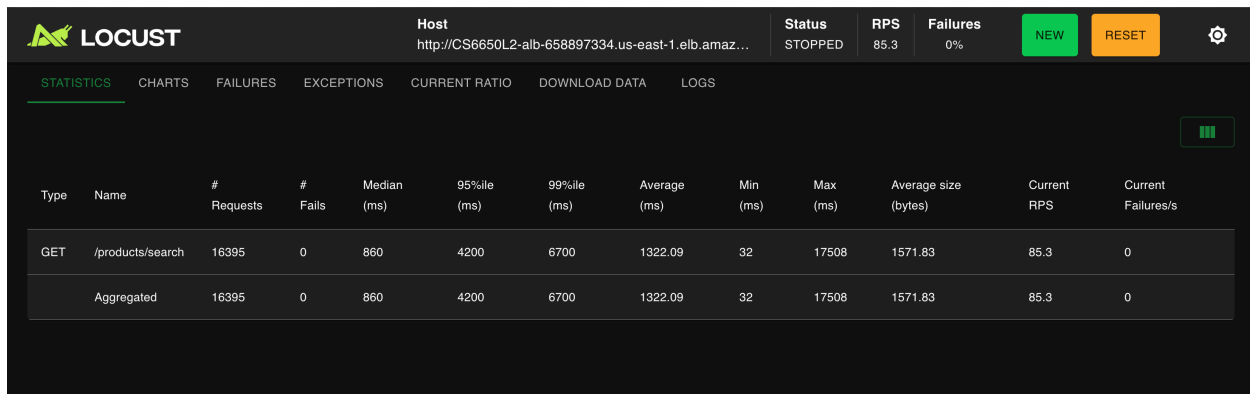
50 users and 50 spawn rate

RPS- 62.5; Median- 660ms; 95th Percentile- 2200ms; 99th Percentile- 3200ms



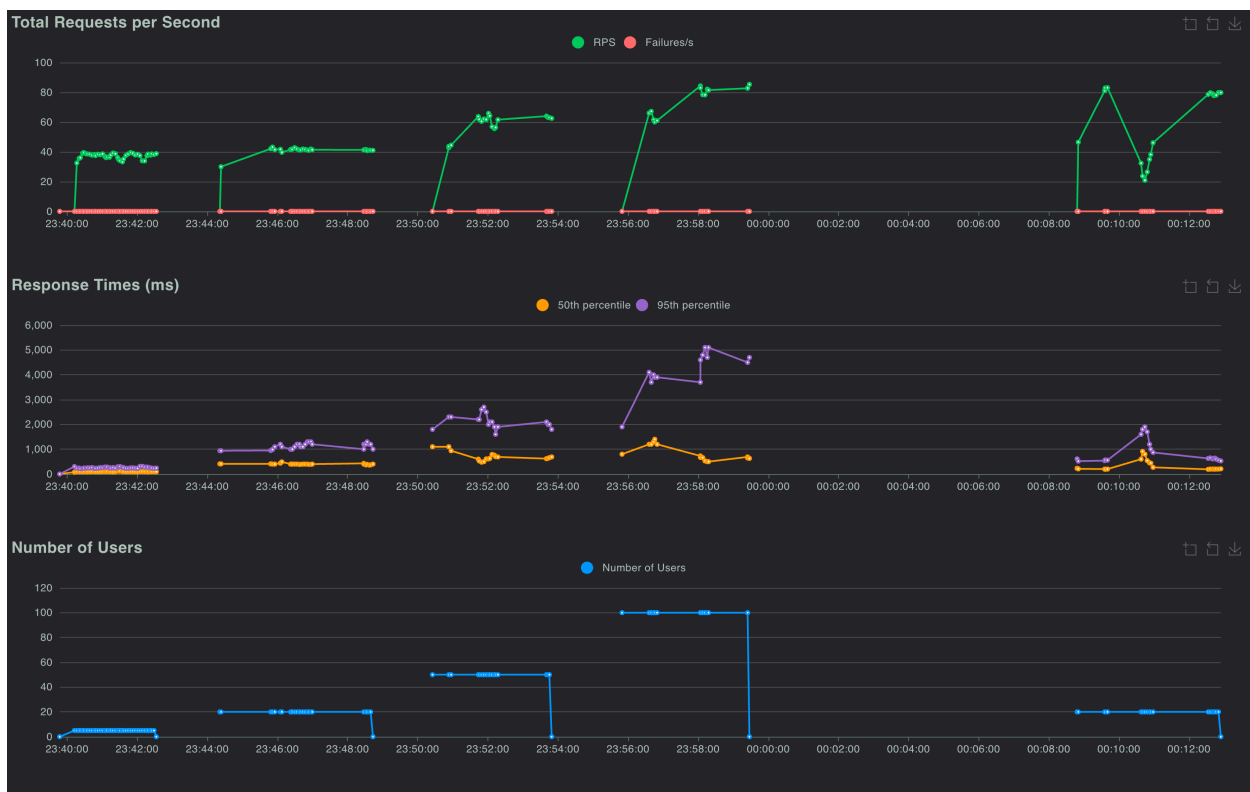
100 users and 100 spawn rate:

RPS- 85.3; Median- 860ms; 95th Percentile- 4200ms; 99th Percentile- 6700ms



Resilience testing:

Three tasks were manually stopped mid-test. 17,607 total requests, 0 failures → ALB instantly rerouted traffic to remaining healthy instances with no service interruption.



Services

Draining

Active

Pending

Running

Tasks

8

23 February 2026, 00:09 (UTC-5:00)

Manage tags

Stop

Run new task

Filter tasks by property or value

Filter desired status

Any desired status

Filter launch type

Any launch type

Task	Last status	Desired status	Task definition	Health status	Created at
337b73dd226949c19387...	Running	Running	CS6650L2-task:3	Unknown	13 minutes ago
4d877c9a412e421c8b17...	Running	Running	CS6650L2-task:3	Unknown	41 minutes ago
7ca1380645f44aa9a4c98...	Running	Running	CS6650L2-task:3	Unknown	19 minutes ago
9b4bd1051c224c19904e...	Running	Running	CS6650L2-task:3	Unknown	41 minutes ago

Services

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

10

23 February 2026, 00:10 (UTC-5:00)

Manage tags

Stop

Run new task

Filter tasks by property or value

Filter desired status

Any desired status

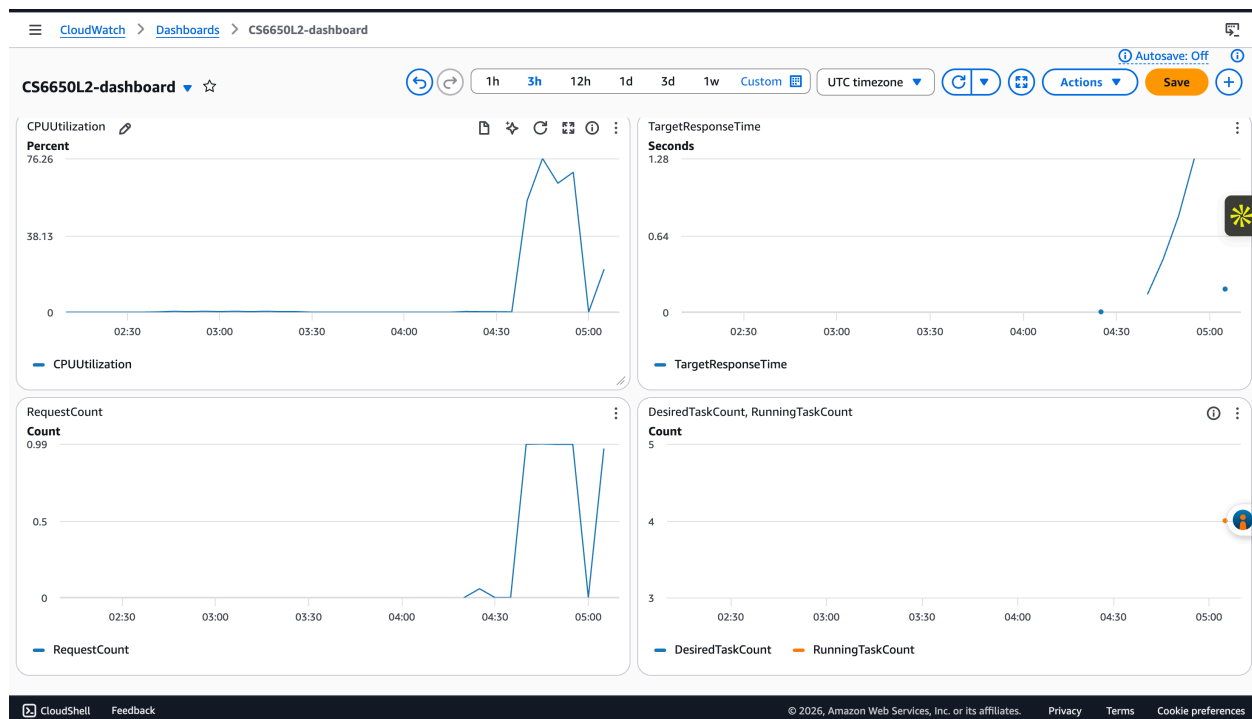
Filter launch type

Any launch type

Task	Last status	Desired status	Task definition	Health status	Created at
337b73dd226949c19387...	Running	Running	CS6650L2-task:3	Unknown	14 minutes ago

CloudWatch dashboard

Four key metrics together → CPU spiked to 76%, TargetResponseTime peaked at 1.28s, RequestCount spike confirms load test, DesiredTaskCount/RunningTaskCount shows scale from 1→4 tasks. This dashboard provides all the evidence needed for scaling decisions.



Comparison table:

Users	RPS (1 task)	Median (1 task)	p95 (1 task)	p99 (1 task)	RPS (ALB)	Median (ALB)	p95 (ALB)	p99 (ALB)
5	20.9	200ms	490ms	600ms	38.8	96ms	260ms	320ms
20	25.7	880ms	1800ms	2500ms	41	410ms	1100ms	1500ms
50	21.4	2100ms	5000ms	6800ms	62.5	660ms	2200ms	3200ms
100	~22	~4000ms	~12000ms	-	85.3	860ms	4200ms	6700ms